

Surface Lattice Resonances in Self-Assembled Gold Nanoparticle Arrays: Impact of Lattice Period, Structural Disorder, and Refractive Index on Resonance Quality

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Cite This: <https://dx.doi.org/10.1021/acs.langmuir.0c02430>



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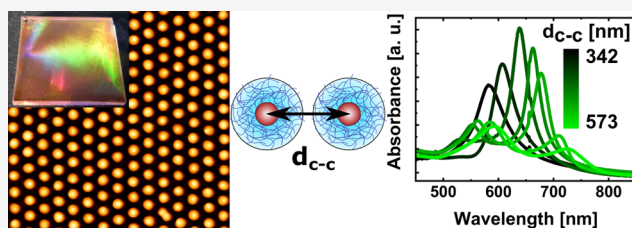


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ABSTRACT: Surface lattice resonances are optical resonances composed of hybridized plasmonic and diffractive modes. These collective resonances occur in periodic arrays of plasmonic nanoparticles with wavelength-scale interparticle distances. The appearance and strength of surface lattice resonances strongly depend on the single particle localized surface plasmon resonance and its spectral overlap with the diffractive modes of the array. Coupling to in-plane orders of diffraction is also strongly affected by the refractive index environment and its symmetry. In this work, we address the impact of the interparticle distance, the symmetry of the refractive index environment, and structural imperfections in self-assembled colloidal monolayers on the plasmonic–diffractive coupling. For this purpose, we prepared hexagonally ordered, nonclose packed monolayers of gold nanoparticles using a fast and efficient, interface-mediated, colloidal self-assembly approach. By tuning the thickness and deformability of the polymer shells, we were able to prepare monolayers with a broad range of interparticle distances. The optical properties of the samples were studied experimentally by UV–Vis spectroscopy and theoretically by finite difference time domain simulations. The measured and simulated spectra allow a comprehensive analysis of the details of electromagnetic coupling in periodic plasmonic arrays. In particular, we identify relevant criteria required for surface lattice resonances in the visible wavelength range with optimized quality factors in self-assembled monolayers.



INTRODUCTION

Metal nanostructures are known to interact with external electromagnetic fields due to the excitation of surface plasmon modes—collective oscillations of the conduction electrons.¹ The strength and energy as well as the number of these modes depend critically on the type of metal,² the nanostructure size^{3,4} and shape,^{3,5,6} the dielectric properties of the surrounding environment,^{3,7} and the nanostructure arrangement.^{8–13} Individual nanoparticles composed of gold or silver with diameters smaller than approximately 100 nm primarily support single localized surface plasmon resonances (LSPRs) that are due to dipolar excitations of the conduction electrons.^{4,14} These dipolar LSPRs can be excited with light in the visible wavelength range, and this phenomenon explains the vivid colors of dilute dispersions of gold and silver nanoparticles.^{15,16} Meanwhile, for the smallest sizes, up to a few tens of nanometers in diameter, the interaction with light is dominated by absorption, and scattering becomes dominant for larger sizes because (Rayleigh) scattering intensity scales with the sixth power of the particle diameter ($I_{sc} \propto D^6$). LSPRs concentrate electromagnetic fields into nanoscale volumes and exhibit extinction cross sections exceeding those of molecular absorbers such as organic dyes by several orders of magnitude, and consequently, they may have applications in solar energy harvesting,^{17,18} ultrasensitive sensing,¹⁹ lasing,²⁰ and photo-

catalysis.²¹ Nevertheless, gold, in particular, is a relatively poor plasmonic material and suffers high intrinsic absorption losses in the visible region, in particular, at shorter wavelengths.^{22,23} The full width at half-maximum (FWHM) of the LSPR is approximately 80–100 nm for gold nanoparticles in a size range of ~8–100 nm.^{24,25} The strong damping and the resulting low quality factors ($Q = \lambda/\Delta\lambda$) limit the potential of gold nanoparticles for many nanophotonic applications. Consequently, strategies are needed to enhance the Q -factor; ideally, these strategies should employ both simple building blocks and scalable fabrication procedures. A powerful strategy for reducing losses in noble metal nanostructures is to arrange the nanoparticles into periodic arrays with wavelength-scale periodicities. Short- and long-range interactions in such arrays can result in higher field enhancements and resonances with significantly improved Q -factors.^{26,27} For large enough periodicities, spectrally narrow surface lattice resonances (SLRs) result from the collective radiative coupling between

Received: August 18, 2020

Revised: October 19, 2020

the single particle LSPRs and diffractive modes. In the case of coupling to in-plane modes of diffraction, i.e., the grating Rayleigh anomaly (RA),²⁸ a uniform dielectric environment surrounding the plasmonic particle array is required.^{24,29} It has been shown that an asymmetric environment suppresses long-range interactions between the nanoparticles due to scattering losses.³⁰ For an in-plane homogeneous environment, the RAs lead to the formation of propagating grazing waves in the plane of the array.³¹ SLRs were first observed and described in 2008 by Kravets *et al.*³² and by Auguie and Barnes.³³ Since then, fundamental spectroscopy of SLRs and the experimental realization of periodic plasmonic arrays that can support high Q-factor SLRs have both become topics of intense studies, as reflected by the appearance of several recent reviews.^{9,24,34}

Until now, electron-beam lithography has been the method of choice for producing periodic particle arrays with nearly perfect structural control. Using this method, arrays with different lattice symmetries³⁵ and even arrays comprising asymmetric particle dimers³⁶ have been prepared. However, more recently, methods suitable for large-area fabrication such as soft interference lithography have gained increasing interest.³⁷ As an alternative, our group has demonstrated that colloidal self-assembly of core-shell building blocks is well suited to the production of periodic plasmonic particles arrays that can extend over cm² areas.^{38–40} We have also shown that such arrays can support the formation of SLRs, albeit possessing relatively low Q-factors.^{41,42} Although self-assembly offers rapid, low-cost, and low-tech production of large-area arrays from relatively simple building blocks,¹⁰ the method typically results in significantly more defects and smaller domain sizes. Hence, it is important to identify experimental parameters that can yield high quality resonances. For domain sizes much larger than the exciting wavelength, a larger lattice period will result in a higher quality mode;^{33,43} however, different rules may apply to self-assembled lattices with limited domain size and multidomain character. A direct comparison between experimental and theoretical extinction properties of imperfect periodic particle arrays as a function of lattice period is currently not available. Consequently, it remains unclear how competitive self-assembly strategies are compared to other techniques used for the fabrication of particle arrays that support SLRs.

In this work, we investigate the optical properties of periodic plasmonic monolayers in both symmetric and asymmetric refractive index environments. We prepare these monolayers by a readily available, robust, and efficient interface-mediated self-assembly procedure that yields hexagonally ordered particle arrangements on cm² substrates. By using colloidal building blocks with plasmonic gold nanoparticle cores and soft, deformable polymer shells of different thicknesses, we obtain lattices with a broad range of interparticle distances. This allows us to study the optical response of the periodic structures and the effects of spectral overlap between the single particle LSPRs and the in-plane modes of diffraction. Different electromagnetic coupling scenarios are identified including long-range (radiative) coupling to grating RAs. We study the effects of limited domain size, scattering losses at domain boundaries, and other structural imperfections. Our findings enable the Q-factor of surface lattice resonances in self-assembled periodic plasmonic monolayers to be optimized.

EXPERIMENTAL SECTION

Chemicals. Gold(III) chloride trihydrate (HAuCl₄; Sigma-Aldrich, ≥99.999%), sodium citrate dihydrate (Sigma-Aldrich, ≥99%), sodium dodecyl sulfate (SDS; Sigma-Aldrich, p.a.), butenylamine hydrochloride (BA; Sigma-Aldrich, 97%), *N*-isopropylacrylamide (NIPAM; TCI, >98.0%), *N,N'*-methylenebisacrylamide (BIS; Sigma-Aldrich, ≥99%), potassium peroxydisulfate (PPS; Sigma-Aldrich, ≥99%), cetyltrimethylammonium chloride (CTAC; Sigma-Aldrich, 25 wt %, H₂O), ascorbic acid (Roth, p.a.), ethanol (Honeywell, p. a.), and 1,4-dioxane (p.a., Fisher Chemicals) were used as received. Milli-Q water (Millipore, R > 18 MΩcm) was used for all syntheses, purifications, and self-assembly steps.

Synthesis. Core-shell particles with polycrystalline gold cores and soft hydrogel shells (Au-PNIPAM) were synthesized in different steps. First, spherical, citrate-stabilized gold nanoparticles (Au-NPs) of approximately 15 nm in diameter were synthesized according to the Turkevich protocol.⁴⁴ Afterward, seeded precipitation polymerization of NIPAM and the cross-linker BIS using the Au-NPs as seed particles was performed following the protocol by Rauh *et al.*⁴⁵ Using three different NIPAM feed concentrations and fixed cross-linker contents (15 mol % referred to NIPAM), we prepared core-shell particles with different shell sizes (see Table 1). Finally, the gold cores of all Au-

Table 1. Concentration (*c*) and Masses (*m*) of the Monomer (NIPAM) and Cross-Linker (BIS) Used for the Seeded Precipitation Polymerization^a

sample	<i>c</i> (NIPAM) [M]	<i>m</i> (NIPAM) [g]	<i>m</i> (BIS) [g]	<i>d_h</i> [nm]
Au-PNIPAM ₃₃₆	0.052	0.587	0.134	336
Au-PNIPAM ₃₅₄	0.058	0.656	0.134	354
Au-PNIPAM ₃₆₇	0.084	0.946	0.194	367

^aThe reaction volumes were 100 ml for each synthesis. Hydrodynamic diameter (*d_h*) of the core-shell particles from dynamic light scattering measured in dilute aqueous dispersion in the swollen state of the shell (25 °C).

PNIPAM batches were stepwise overgrown with gold to reach approximately 100 nm in diameter. The overgrowth protocol was adapted from Honold *et al.*³⁸ For the sake of clarity, the different core-shell samples are labeled Au-PNIPAM₃₃₆, Au-PNIPAM₃₅₄, and Au-PNIPAM₃₆₇ with the numbers in subscripts corresponding to the hydrodynamic radius of the core-shell particles measured at 25 °C. The final particles were purified by centrifugation and redispersion of the residues in water. Centrifugation and redispersion were repeated three times at 5200 rcf (relative centrifugal force) for 30 min each. After the last centrifugation, the entire batch particles was redispersed in 150 μL of ethanol. A detailed description of the synthesis steps as well as exemplary TEM images and UV-Vis absorbance spectra from each synthesis step (Figure S1) can be found in the Supporting Information.

Fabrication of Periodic Particle Arrays. The fabrication of hexagonally ordered Au-PNIPAM particle arrays on glass substrates was carried out by following the protocol of Volk *et al.*⁴¹ Briefly, a small crystallization dish was filled with Milli-Q water and 7 μL of the particle dispersion (purified, in ethanol) was gently transferred to the air/water interface. At the interface, the particles spontaneously self-assembled into a freely floating monolayer. Floating monolayers were transferred onto glass substrates by immersing the substrate fully in the bulk water phase below the monolayer and then withdrawing the substrate slowly through the monolayer. Therefore, the monolayer was collected onto the glass. Immediately after withdrawal of the substrates, the samples were dried using a heat-gun to blow warm air onto the back of the glass. A video demonstrating the self-assembly process can be found in the Supporting Information. To achieve the homogeneous refractive index environment required for radiative coupling to in-plane modes of diffraction, the monolayer samples were coated in a PNIPAM layer by spin-coating linear PNIPAM (65,000 g/

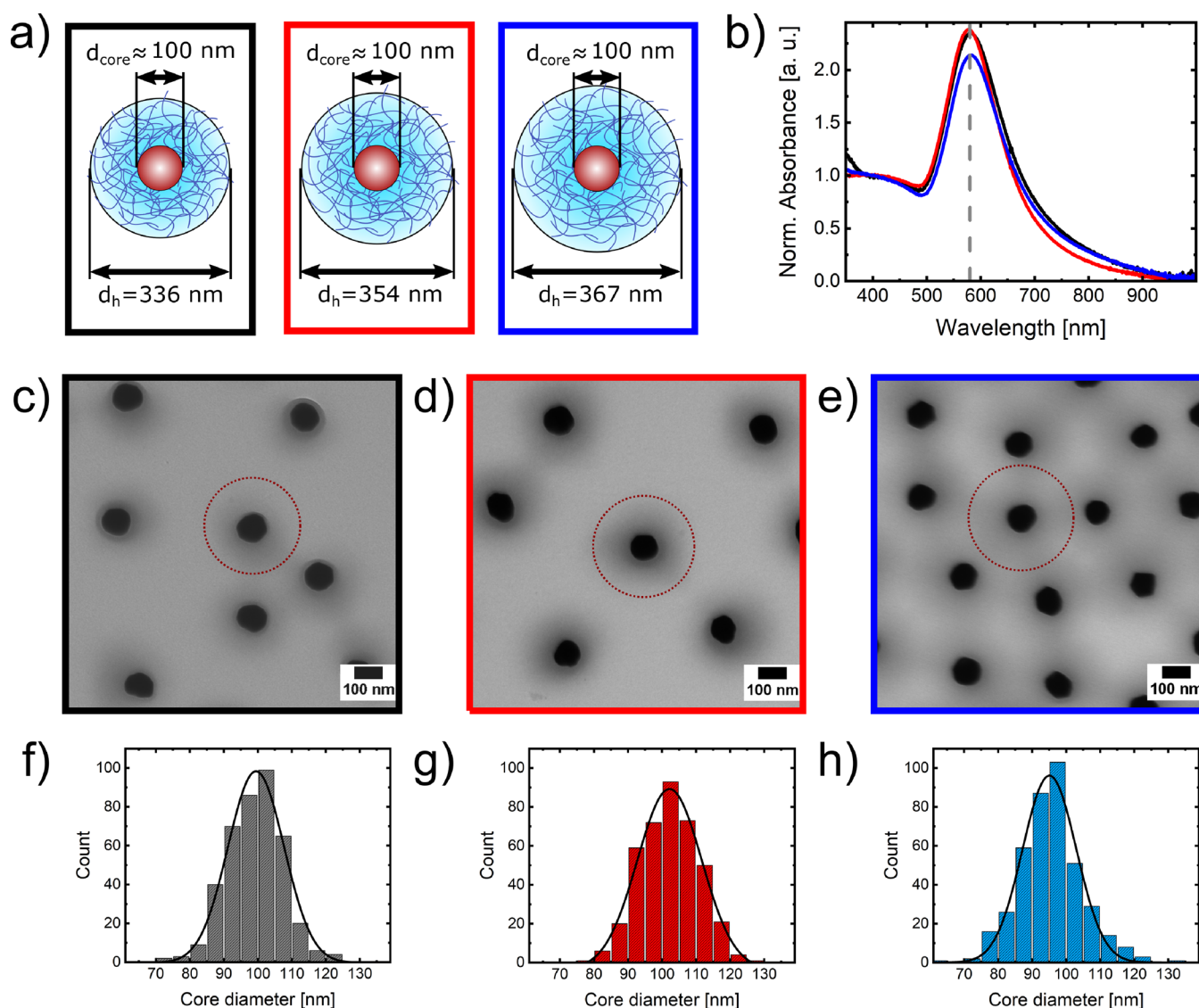


Figure 1. Au-PNIPAM building blocks for monolayer preparation: Au-PNIPAM₃₃₆ in black, Au-PNIPAM₃₅₄ in red, and Au-PNIPAM₃₆₇ in blue. (a) Schematic representation of the core–shell structure of the Au-PNIPAM particles with d_{core} indicating the core diameter and d_h corresponding to the hydrodynamic diameter. (b) Absorbance spectra normalized to the absorbance at 400 nm. The dashed gray line highlights the LSPR position (λ_{LSPR}). (c–e) Representative TEM images for each batch of particles. The red dotted circles indicate the corresponding hydrodynamic size determined from DLS. (f–h) Histograms for the core diameters determined from TEM images.

mol, synthesized according to Ebeling and Vana⁴⁶). Spin-coating was performed by applying 60 μL of a 1,4-dioxane solution containing the polymer (100 mg of PNIPAM in 2 mL of dioxane) and then applying 2000 rpm for 90 s. Under these conditions, the resulting PNIPAM films had thicknesses of approximately 300 nm.

Methods. Transmission Electron Microscopy. A Zeiss CEM902 transmission electron microscope (TEM) was used for the determination of particle size, size dispersity, and structure. TEM measurements were performed at an acceleration voltage of 80 kV in bright-field mode. For each sample, a drop of a dilute aqueous particle dispersion was placed on carbon-coated copper grids and left to dry in air. Statistical analysis of the core sizes was carried out by measuring 200 particles from various TEM images using the ImageJ software.⁴⁷

Atomic Force Microscopy. Atomic force microscopy (AFM) topographic images ($10 \times 10 \mu\text{m}^2$) were recorded with a Nanowizard 4 (JPK Instruments) in intermittent contact mode against air. OTESPA-R3 AFM probes (Bruker) were employed for image recording. The cantilevers possess a resonance frequency of approximately 300 kHz and a spring constant of approximately 26

N/m. The tip geometry was a visible apex with a nominal tip radius of 7 nm.

UV–Vis Spectroscopy. The extinction spectra were recorded with a Specord S 600 UV–Vis spectrophotometer (Analytik Jena AG) in transmission geometry. The spectra of the particle dispersions were measured in $1 \times 1 \text{ cm}^2$ PMMA cuvettes. Particle monolayers on glass substrates were positioned upright in the light path using a solid sample holder. Thus, the illumination is normal to the monolayer plane. The transmitted light is detected and converted into absorbance values. The measurement geometry is also depicted in a sketch in the Supporting Information (Figure S8). All extinction spectra were corrected for background using either a cuvette with water or a clean glass substrate.

Dark-Field Microscopy. The optical microscopy images were captured in dark-field mode with reflection configuration using a Nikon ECLIPSE LV100D-U microscope equipped with a halogen lamp (12 V, 50 W), a dark-field condenser, and a CMOS camera (Thorlabs, DCC1645c-HQ). The microscope was equipped with three lenses, Nikon LU Plan Fluor 20x/0.45, Nikon LU Plan ELWD 50x/0.55, and Nikon Lu Plan Fluor 100x/0.9. The local trans-

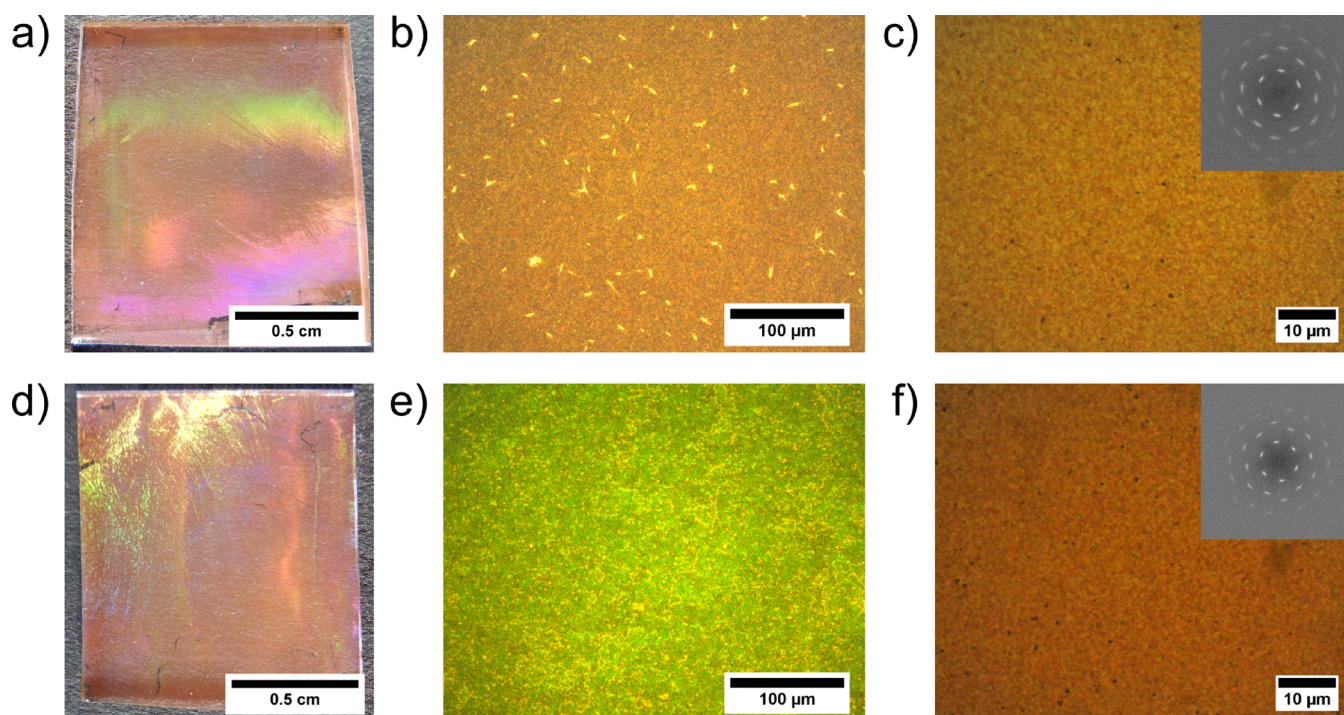


Figure 2. Monolayer characterization at different magnifications. Top row: Monolayer with $d_c - c = 458 \pm 29$ nm. Bottom row: $d_c - c = 567 \pm 32$ nm. (a, d) Digital photographs of the samples. (b, e) DFM images with 20X magnification. (c, f) DFM images with 100X magnification. The insets show FFTs calculated from the DFM images.

mission spectra were recorded with a CCD camera (Princeton Instruments, Pixis 1024F) and an imaging spectrograph (Princeton Instruments, Acton MicroSpec 2150i).

FDTD Simulations. The extinction spectra and E-field maps were calculated using the FDTD (finite difference time domain) method with commercial software from Lumerical Solutions, Inc. (FDTD Solutions, Version 8.18.1332). For the finite size lattices, the absorption and scattering spectra were simulated in a box with perfectly matched layer boundary conditions in both the x - and y -directions (layer direction) and the z -direction. Unless otherwise specified, the finite size lattices consisted of 61 particles, which corresponded to a domain size of four. We utilized a total-field scattered-field source and a box of power monitors consisting of the total field and scattered field monitors. For the infinite lattices and electromagnetic field simulations, periodic boundary conditions in both the x - and y -directions (layer direction) and a perfectly matched layer in the z -direction (beam direction) were used. A linear polarized plane wave source with excitation along the z -axis was injected. A transmission monitor was located at both ends of the FDTD simulation box in the beam direction. The following settings were kept the same for all simulations. Particle sizes and distances were used as obtained from the experiments. The wavelength dependent RI (refractive index) of the gold core was taken from literature.⁴⁸ For the glass substrates, a constant RI of 1.52 was used and for the air background 1. The PNIPAM film was simulated with 300 nm thickness and a constant RI of 1.49⁴⁹ slightly smaller than 1.50 (at $\lambda = 600$ nm)⁵⁰ to account for a small amount of residual water. The auto shut-off was set at 10^{-5} . A schematic depiction of the simulation geometry is shown in the Supporting Information (Figure S8).

RESULTS AND DISCUSSION

Au-PNIPAM Particle Characterization. Core-shell colloids with polycrystalline plasmonic gold cores and three thicknesses of soft PNIPAM hydrogel shells were synthesized. A schematic representation of the particles is shown in Figure 1a. We aimed at synthesizing particles that feature nearly the same core diameter (d_{core}) so that the plasmonic properties,

i.e., the position and intensity of the LSPR, are comparable despite the different thicknesses of the dielectric PNIPAM shells. Figure 1b shows the normalized absorbance spectra recorded from dilute aqueous dispersions of the Au-PNIPAM particles. All samples feature a single LSPR peak attributed to a dipolar mode. The LSPR positions are $\lambda_{\text{LSPR}} = 577$ nm for Au-PNIPAM₃₃₆, $\lambda_{\text{LSPR}} = 582$ nm for Au-PNIPAM₃₅₄, and $\lambda_{\text{LSPR}} = 583$ nm for Au-PNIPAM₃₆₇, thus matching closely. Due to the relatively large linewidths (FWHM ≈ 137 nm), all particles possess low Q -factors of $Q \approx 4$. Figure 1c–e shows TEM images of the different particles. In all the three images, the core-shell structure is clearly visible due to the pronounced contrast difference between the high electron density gold cores and the low electron density PNIPAM shells. The shells were imaged in their dried and collapsed state because of the measuring conditions during TEM imaging. Thus, the shells appear thinner than in their swollen state in aqueous dispersion. Dynamic light scattering from dilute dispersions measured at 25 °C (swollen state) revealed hydrodynamic diameters of $d_h = 336$ nm, $d_h = 354$ nm, and $d_h = 367$ nm. The average core sizes (d_{core}) were determined directly from TEM images. The cores of the three core-shell systems have nearly the same size ($d_{\text{core}} = 99 \pm 8$ nm (Au-PNIPAM₃₃₆), $d_{\text{core}} = 102 \pm 8$ nm (Au-PNIPAM₃₅₄), and $d_{\text{core}} = 96 \pm 9$ nm (Au-PNIPAM₃₆₇)). The corresponding size distributions are shown in Figure 1f–h. The standard deviations of the average diameter are 8–9%, emphasizing the good control of our overgrowth procedure (see the Experimental Section and Supporting Information for more details). A further reduction in the polydispersity of the core sizes might be achievable by further optimization of the synthesis protocol, but this is not expected to cause any significant increase in the Q -factor of the LSPR. Indeed, a further increase in core diameter will eventually lead to broader linewidths and thus reduced Q

values due to the appearance of quadrupolar modes and retardation of the dipolar mode.²⁵

Monolayers from Au-PNIPAM Particles. We used the different Au-PNIPAM core-shell particles to prepare hexagonally ordered monolayers with a broad range of nearest-neighbor center-to-center interparticle distances (d_{c-c}) by an interface-mediated self-assembly approach.⁴¹ Control of d_{c-c} was realized (1) by using different shell thicknesses and (2) by using sequential transfer from floating monolayers at the air/water interface where the packing fraction is reduced after every withdrawal of a monolayer sample. A densely packed monolayer is first prepared at the interface. In this case, the value of d_{c-c} is close to the bulk hydrodynamic diameter of the particles.³⁹ Then, a fraction of the monolayer is removed from the interface by transferring onto the first substrate, resulting in an expansion of the remaining monolayer driven by interfacial tension and the deformability of the PNIPAM shells. Now, withdrawal of some of this monolayer onto another substrate yields a sample with a slightly reduced particle density, and d_{c-c} is increased by about 1.02 up to 1.1. Using this multistep transfer process in combination with particles having a range of different shell thicknesses allows the fabrication of monolayers with various values of d_{c-c} . In this work, d_{c-c} was varied between 342 and 448 nm using the Au-PNIPAM₃₃₆ particles, between 448 and 500 nm using the Au-PNIPAM₃₅₄ particles, and between 500 and 573 nm with the Au-PNIPAM₃₆₇ particles. A video of the assembly approach is provided in the [Supporting Information](#) where changes in the monolayer density become clearly visible because the reflected color changes due to the changing particle spacing. Since the spacing in the monolayers is only determined by the dielectric PNIPAM shells and because the sizes and thus LSPR properties of the cores are nearly identical, the only optically relevant parameter that is altered between the samples is the lattice spacing.

Figure 2a,d shows the photographs of the light reflection from particle arrays with different lattice spacings on glass substrates illuminated by a white light source from the side. In both cases, the monolayers cover the full, macroscopic substrates, as evident from the opalescent colors. Additionally, images of the same samples were taken using a dark-field microscope (DFM) at different magnifications to look at the largest possible area of the monolayers. At 20 $\times$ magnification (**Figure 2b,e**), the two samples show clear differences in color. The sample with $d_{c-c} = 458 \pm 29$ nm in **Figure 2b** appears orange, while the sample with the larger lattice spacing ($d_{c-c} = 567 \pm 32$ nm) appears green. The observed colors are a combination of the scattering from the individual particles and the Bragg diffraction. The latter depends strongly on the distance between the particles, the light incident, and the detection angle. At 100 $\times$ magnification, both samples appear similar in color, yellowish-orange caused by the scattering of individual gold particles. Due to the larger aperture of the objective, the diffraction of light by the lattices cannot be observed. The DFM images confirm the homogeneity of the samples on different length scales. Furthermore, the images taken at the highest magnification allow for identification of particle positions at the single particle level. The DFM images recorded at 100 $\times$ magnifications are also shown in the [Supporting Information](#) with larger image sizes (see **Figure S2**). Using the particle positions, we calculated the fast Fourier transformations (FFTs) that are shown as insets in **Figure 2c,f** to evaluate the lattice order on statistically relevant areas with

nearly mm² size. Both FFTs show Bragg peaks with hexagonal symmetry and at least four to five diffraction orders, reflecting the high degree of hexagonal ordering. Thus, on average, our samples feature domain sizes of at least three (37 particles) to four (61 particles). We will later show that indeed we get very good agreement between experimental extinction spectra and FDTD simulation results based on a domain size of four.

Since the resolution of the DFM image in **Figure 2f** is sufficient to track the positions of the large majority of the particles at the single particle level, we can use this image to analyze the crystalline domains. Therefore, we have used triangulation analysis introduced by Hillebrand *et al.*⁵¹ **Figure 3**

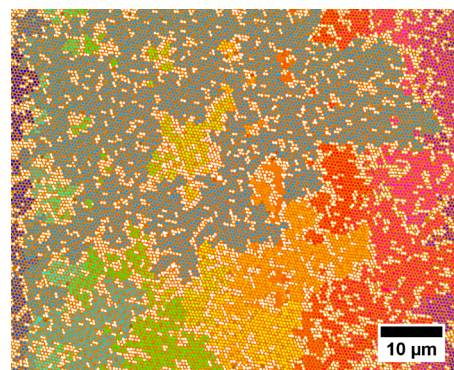


Figure 3. Analysis of crystalline domains. DFM image with $d_{c-c} = 567 \pm 32$ nm. Different crystalline domains are highlighted in different colors. Particles without hexagonal coordination are colored in white.

shows the DFM image with the detected single crystalline domains (hexagonal order) highlighted in different colors. Particles without hexagonal coordination are highlighted in white. At first glance, we see that the majority of the particles have hexagonal coordination and thus are assigned to crystalline domains, which is in good agreement with the FFT results. The image contains a rather large number of domains of various sizes. The largest domain contains more than 5000 particles. Furthermore, we are able to identify 16 domains that contain 100 or more particles.

In order to look more closely at the microstructures and the crystalline domains, we employed AFM to study monolayers with different values of d_{c-c} . Since here we look at smaller sample areas, structural deviations between different sample locations are expected. Overall, the AFM results will be less statistically relevant but still provide important insights into the lattice structure. Representative images are shown in **Figure S3** ([Supporting Information](#)). The images and the respective FFTs of the particle positions confirm the hexagonal ordering of the samples. Furthermore, pair correlation functions $g(r)$ were calculated, allowing quantitative analysis of the degree of lattice order on smaller length scales than the DFM data allow. For all selected samples, $g(r)$ shows at least six to seven clearly distinguishable peaks, indicating large crystalline domain sizes. Throughout this manuscript, the values of d_{c-c} have been calculated from the first maximum of $g(r)$, which has in turn been determined from a Lorentz fit to the function. Since the AFM images still capture several hundreds to thousands of particles, the values of d_{c-c} are representative of the whole monolayer and the error is expected to be small. In order to analyze the domain sizes of monolayers with smaller lattice constants where DFM cannot resolve individual particles, we

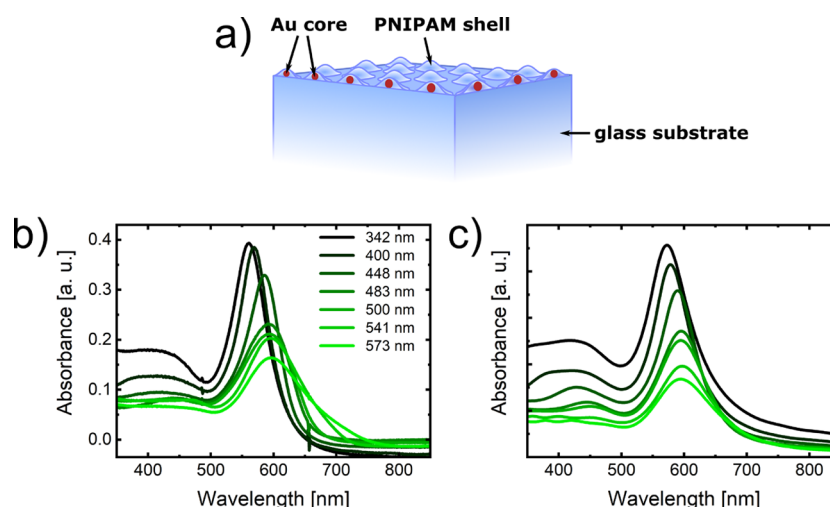


Figure 4. Optical properties of particle arrays in an asymmetric RI environment. (a) Schematic depiction of the periodic particle array on a glass substrate with air as a superstrate. (b) Experimental absorbance spectra measured from samples with values of d_{c-c} ranging from 342 to 573 nm. (c) Corresponding theoretical spectra from FDTD simulations of finite lattices (domain size four) using the experimental values of d_{c-c} from (b). The same color code was used as in (b).

have recorded height profiles by AFM. Figure S4a in the Supporting Information shows the $40 \times 40 \mu\text{m}^2$ scans of monolayer samples with values of d_{c-c} of 375, 458, and 567 nm. From the domain analysis in Figure S4b,c, we find (1) that much larger, single crystalline domains are detected as compared to the previously discussed DFM image and (2) that the domain size seems to increase with increasing d_{c-c} . We can certainly conclude that the degree of order does not decrease with increasing d_{c-c} . This will become relevant when discussing diffractive–plasmonic coupling later on. The fact that larger domains are found from analysis of the AFM images is simply related to the better spatial resolution in AFM. Thus, AFM allows a much more precise determination of the central x - and y -positions of individual particles.

Optical Properties in an Asymmetric RI Environment.

In the following, we will address the dependence of the optical properties of our periodic particle arrays on the interparticle spacing (d_{c-c}). We start by considering arrays where there is a glass substrate (RI = 1.52) and air (RI = 1.00) is the superstrate. A schematic depiction of the sample geometry is shown in Figure 4a. Figure 4b summarizes the absorbance spectra of selected samples measured in normal transmission. The spectra from a larger set of monolayers can be found in the Supporting Information (Figure S5a). All of these arrays exhibit a single resonance peak whose resonance position, width, and intensity depend on the lattice spacing. With decreasing d_{c-c} , the resonance shifts to smaller wavelengths, becoming more intense, and the FWHM decreases. The increase in resonance intensity is primarily attributed to the increasing particle density (Beer–Lambert law) as the lattice spacing is reduced. For the largest spacing, where $d_{c-c} = 573$ nm, we find a resonance position $\lambda_{\text{max}} = 595$ nm and a FWHM of 145 nm. Both values are close to the ones obtained from measurements in dilute aqueous dispersions (RI = 1.33) with $\lambda_{\text{max}} \approx 580$ nm and a FWHM of 137 nm. The slight redshift of the monolayer sample is related to the small increase in the effective RI for the combination of the glass substrate with the air superstrate. There is also a small increase in the effective RI due to the collapsed PNIPAM shell surrounding each gold core. We have also measured the spectrum of a monolayer

sample on glass with a random particle distribution and a very small packing fraction (see Figure S6). This yields a resonance at $\lambda_{\text{max}} = 595$ nm, perfectly matching the resonance of the ordered monolayer sample with $d_{c-c} = 573$ nm. This confirms that electromagnetic coupling is absent for large lattice spacings and the observed resonance shifts compared to particles in solution are due to changes in RI environment. If we now look at the monolayer sample with the smallest lattice spacing ($d_{c-c} = 342$ nm), the resonance peak appears at $\lambda_{\text{max}} = 561$ nm, i.e., it is significantly blueshifted compared to the LSPR observed in dilute particle dispersions. Furthermore, the FWHM is reduced to 78 nm. Both the blueshift and the reduction in linewidth are attributed to long-range radiative interactions between the gold nanoparticle dipoles.^{52,53} At these still relatively large interparticle distances, we can exclude near-field electromagnetic coupling. Near-field coupling becomes significant for interparticle spacings smaller than 2.5 times the particle diameter in the case of gold.⁵⁴ Furthermore, coupling via the near-fields is expected to result in a redshift and resonance broadening in stark contrast to our observation. Thus, our resonance effect is clearly radiative (far-field). In our case, the interparticle distances are on the order of half the LSPR wavelength. Such blueshifts have been observed in similar periodic arrays and attributed to radiative dipole interactions and retardation effects.⁵² The blueshift and narrowing with decreasing distance have also been observed in random particle monolayers.⁵³ This suggests that the long-range radiative coupling is mostly determined by the average interparticle distance rather than the arrangement of the plasmonic building blocks. Unfortunately, a detailed description of the physical origin of the blueshift, for example, whether this is an effect related to changes in the scattering or absorption properties of the particle ensemble, is still missing until now.

In order to support our experimental findings, we have also performed FDTD simulations. The calculated spectra are shown in Figure 4c. The observed peak positions, widths, and intensities nicely match our experimental results, as is seen from a direct comparison of the corresponding values summarized in Table 2.

Table 2. Calculated Wavelength of the {1,0} Diffraction Peak, λ_{Bragg} , Resonance Positions, λ_{max} , and FWHM Values for Lattices with Various d_{c-c} for Both Asymmetric and Symmetric RI Environments^a

d_{c-c} [nm]	λ_{Bragg} [nm]	λ_{max} [nm]		FWHM [nm]	
		asymmetric	symmetric	asymmetric	symmetric
342 ± 44	441	561 (572)	583 (588)	78 (90)	75 (84)
400 ± 52	516	569 (578)	607 (612)	71 (77)	52 (65)
448 ± 60	578	585 (590)	638 (642)	69 (79)	40 (47)
483 ± 46	623	592 (596)	662 (670)	106 (103)	33 (50)
500 ± 51	645	593 (596)	678 (684)	127 (111)	42 (61)
541 ± 30	698	595 (596)	710 (718)	136 (121)	67 (107)
573 ± 42	739	595 (596)	726 (750)	145 (127)	124 (132)

^aValues in brackets correspond to the results from FDTD simulations.

Optical Properties in Symmetric RI Environments. In order to enhance radiative electromagnetic coupling, we need a higher RI superstrate that closely matches the RI of the glass substrate. To achieve this, we have deposited 300 nm-thick polymer coatings on top of the particle arrays. This thickness is sufficient to enable observation of diffractive–plasmonic coupling.⁴² In contrast to other existing strategies for refractive index matching using immersion oils or photoresists, our superstrate is swellable by solvents⁴¹ and will in principle enable the diffusion of small analytes into the film as desired, for example, in sensing applications.⁵⁵ A schematic depiction of the sample geometry is shown in Figure 5a. Note that, first of all, we need to determine the uncoupled single particle resonance as a reference. Figure 5b shows the simulated, single particle spectrum for a Au-PNIPAM particle on a glass substrate with a 300 nm-thick topcoating of PNIPAM (RI = 1.49). We predict an LSPR at $\lambda_{\text{LSPR}} = 604$ nm, highlighted by the vertical gray dashed line. As a consequence of the higher RI

superstrate, the LSPR position is redshifted (by approximately 9 nm). The colored vertical lines in Figure 5b correspond to the calculated spectral positions of the {1,0} diffraction modes for the hexagonally ordered monolayers with varying d_{c-c} . The theoretical wavelengths of the Bragg modes were calculated using eqs 1 and 2⁴²

$$\frac{1}{d^2} = \frac{4}{3} \left(\frac{h^2 + hk + k^2}{a^2} \right) \quad (1)$$

Here, d is the interplanar spacing, h and k are the Miller indices (in our case 1 and 0, respectively), and a is the lattice constant. For our hexagonally ordered monolayers, a corresponds to the nearest-neighbor interparticle distance d_{c-c} . Multiplication of the interplanar spacing d with the refractive index yields the Bragg wavelengths

$$\lambda_{hk} = RI \times d \quad (2)$$

Figure 5c shows the experimental absorbance spectra for selected monolayer samples immersed in symmetric RI environments. The spectra from a larger set of monolayers can be found in the Supporting Information (Figure S5b). To allow for a better comparison of the resonance intensities, the spectra in Figure 5c have been normalized to the same particle density (10 particles per μm^2). Looking at the smallest values of d_{c-c} (342 and 400 nm) where the diffractive modes are significantly blueshifted compared to the single particle LSPR (Figure 5b), we observe single resonance peaks. Here, the sample with $d_{c-c} = 400$ nm features a resonance that is more intense, redshifted, and with a reduced linewidth compared to $d_{c-c} = 342$ nm. As in the case of asymmetric RI environments, we attribute these spectral differences to long-range radiative interactions between the dipoles. At $d_{c-c} = 448$ nm, the strongest resonance appears at 638 nm with an even higher intensity and reduced linewidth. For this sample, the {1,0}

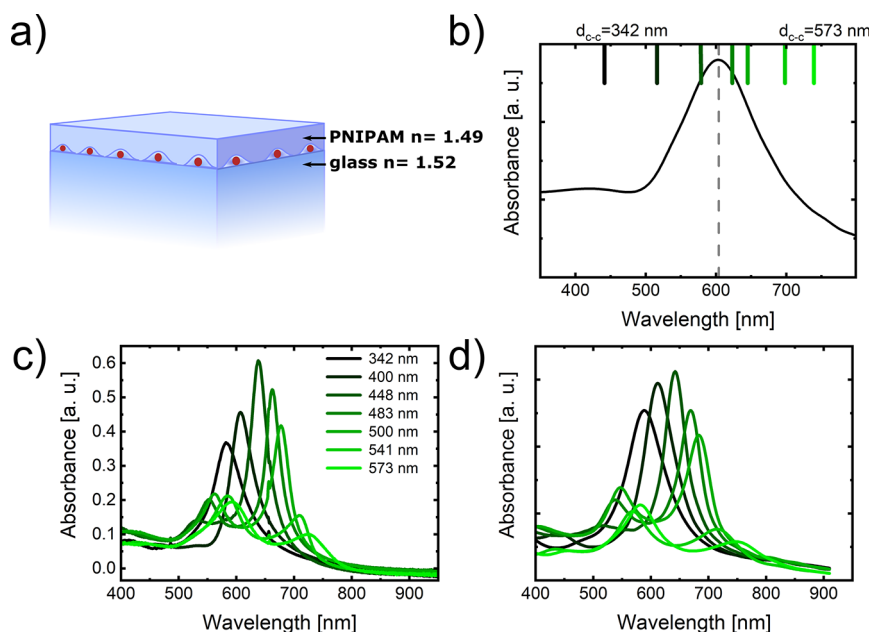


Figure 5. Optical properties of particle arrays in symmetric RI environments. (a) Schematic figure of a PNIPAM-coated gold particle monolayer on the glass substrate. (b) Calculated LSPR of a single gold particle on the glass substrate coated with a 300 nm PNIPAM film with a theoretical Bragg position for each interparticle distance. (c) Absorbance with varying interparticle distance. (d) FDTD simulated absorbance spectra of particle lattices (domain size four) with varying spacing in the homogeneous RI environment. The color code in (b–d) refers to the same d_{c-c} . All spectra were normalized to account for the same number of particles per area (10 particles per μm^2).

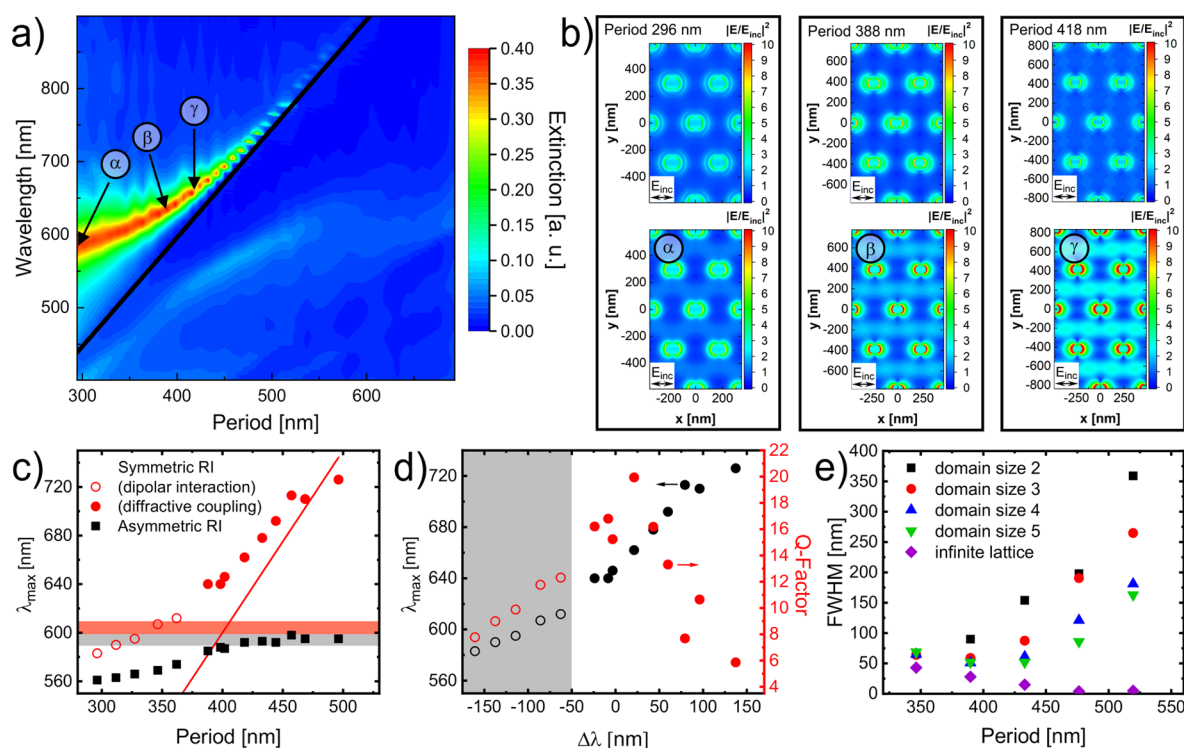


Figure 6. Comparison of different electromagnetic coupling regimes. (a) Calculated extinction cross sections for hexagonal arrays of gold spheres (100 nm diameter) for lattice periods in 10 nm steps and a symmetric environment with RI = 1.49 for the PNIPAM topcoating and RI = 1.52 for the glass substrate. The solid line corresponds to the (0,±1) Rayleigh anomaly (RA). (b) Near-field electromagnetic intensity maps ($|E/E_{\text{inc}}|^2$) from FDTD simulations for a hexagonal monolayer with asymmetric RI (top row) and symmetric RI environments (bottom row). Calculations were performed at the respective λ_{max} of the samples. Corresponding spectral positions for the symmetric case labeled with α , β , and γ are also indicated by arrows in (a). (c) Peak wavelength (λ_{max}) as a function of lattice period from experiments. Black squares: asymmetric RI. Red circles: symmetric RI. Filled red circles correspond to samples where diffractive–plasmonic coupling (SLRs) were observed. The red line corresponds to the (0,±1) RA. The light red and gray bars mark the LSPR position (no coupling) in symmetric and asymmetric RI, respectively. (d) Experimental peak wavelength (λ_{max}) and Q-factor as a function of wavelength difference between LSPR and RA ($\Delta\lambda = \lambda_{\text{RA}} - \lambda_{\text{LSPR}}$). (e) FWHM of resonance peaks from FDTD simulations of hexagonal lattices in symmetric environment (RI = 1.49 for the PNIPAM topcoating, RI = 1.52 for the glass substrate) for different domain sizes. (The associated absorbance spectra from (c) and (d) are shown in Supporting Information Figure S5)

diffraction modes and the single particle LSPR overlap strongly and diffractive–plasmonic coupling is enabled, resulting in a true SLR. Although the RI values of the PNIPAM superstrate (RI = 1.49) and the glass substrate (RI = 1.52) are not perfectly matched, an SLR is supported. This finding is in agreement with the work of Yang *et al.* who showed that SLRs are tolerant of RI mismatches up to $\Delta\text{RI} \approx 0.05$.⁵⁶ In addition, the spectrum features a weak shoulder at approximately 530 nm. This mode is attributed to a weak plasmonic contribution.⁵⁷ With further increases in $d_{\text{c-c}}$, the SLRs redshift and become broader and weaker. At the same time, the plasmonic contribution at shorter wavelengths becomes more pronounced. For the sample with the largest spacing ($d_{\text{c-c}} = 573$ nm), where the {1,0} diffraction modes appear in the long-wavelength tail of the single particle LSPR, the plasmonic contribution almost resembles the uncoupled LSPR response of the component building blocks and only a weak and broad SLR at 726 nm is observed. The increase in linewidth of the SLR with increasing $d_{\text{c-c}}$ indicates a decrease in the coupling strength. Although for a perfect particle lattice with an effectively infinite particle number, a reduction of the SLR linewidth is expected for increasing lattice constants, we clearly observe the opposite effect.

We explain this in terms of the finite size of the single crystalline domains in our monolayers as discussed in the previous section. Our explanation is supported by the work of

Zou *et al.*⁵⁸ Those authors studied the optical response of a two-dimensional array of 400 nanoparticles in a hexagonal arrangement and found an increase in resonance linewidth with increasing spacing. Apart from the influence of the domain size, we also expect the multidomain character of the samples to contribute to the resonance broadening at large lattice constants. The multidomain character and the resulting large number of grain boundaries dampen the diffractive modes. In addition, there will be significant scattering losses at the domain boundaries as observed for guided resonances in 2D photonic crystals.⁵⁹

To further support our experimental findings, we performed FDTD simulations of lattices with finite sizes. Figure Sd shows the calculated spectra using the same lattice parameters as in the experiment and a domain size of four corresponding to 61 particles. Again, the simulated spectra have been normalized to 10 particles per μm^2 . All peak positions as well as the experimental and simulated values of the resonance FWHM are presented in Table 2. We find a very good match between experiments and simulation for all the different monolayers investigated, despite a small deviation in the SLR position that can be explained in terms of a small amount of positional disorder in the self-assembled lattices. This disorder weakens the dipolar coupling in our lattices, resulting in slightly blueshifted SLRs as compared to perfectly ordered lattices used in simulation. SLRs are coherent modes that result from the

collective radiative coupling between individual plasmonic nanoparticles in the periodic lattices. The stronger the coupling and consequently the larger the coherence, the stronger is the redshift of the SLR with respect to the diffraction edge.⁶⁰ In contrast, for randomly arranged plasmonic particles in a monolayer, the SLR is expected to disappear, while the purely plasmonic contribution at lower wavelength completely reappears. This is in agreement with the study of Auguie and Barnes.⁶⁰ Those authors found a blueshift and broadening of SLRs for increasing disorder.

In addition to the far-field extinction spectra, we have also recorded the local transmission spectra using a light microscope. The resulting spectra for the monolayer samples with $d_{c-c} = 449$ nm and $d_{c-c} = 573$ nm in asymmetric and symmetric RI environments are shown in the Supporting Information (Figure S7). The local spectra resemble the spectra collected by standard extinction spectroscopy. For the symmetric RI environment, we observe SLRs for both lattice constants, while the $d_{c-c} = 573$ nm sample exhibited a pronounced plasmonic contribution that almost matches the expected single particle LSPR as well as the noncoupling case in an asymmetric RI environment.

Direct Comparisons of Asymmetric and Symmetric RI Environments. We now want to directly compare and discuss the differences in optical response in asymmetric and symmetric RI environments. In order to be comparable to other studies in the literature and to other lattice symmetries, we will use the lattice period instead of d_{c-c} in the following discussions. Figure 6a shows the extinction cross sections obtained from FDTD simulations for hexagonal infinite lattices of spherical Au-NPs (100 nm in diameter) for various lattice periods in 10 nm steps. A constant RI = 1.49 for the PNIPAM topcoating as a superstrate and RI = 1.52 for the glass substrate were used (symmetric environment), which match closely the expected RIs from the experiment. With increasing period, the resonance maximum continuously redshifts, while the linewidths become smaller. For large periods, the resonance positions asymptotically approach the $(0, \pm 1)$ RA (solid line), implying that the resonance character becomes more and more photonic and less plasmonic. Figure 6b compares the electromagnetic near-field intensities calculated at the respective resonance wavelength ($\lambda_{\max}$) for the asymmetric (top row) and symmetric (bottom row) RI cases for three selected lattice periods. All maps reveal the excitation of dipolar resonances with stronger field strengths for the symmetric RI case caused by the higher effective average RI. In addition to the dipole excitation, the symmetric RI cases with periods of 388 nm ($d_{c-c} = 448$ nm) and 418 nm ($d_{c-c} = 483$ nm) that are labeled with α and β show the typical in-plane diffractive band structure. The diffractive–plasmonic coupling in these two samples results in the formation of SLRs. In all cases the SLRs exhibit the highest field intensities. In experiment, we observe very similar results. Figure 6c shows the resonance wavelength ($\lambda_{\max}$) determined for monolayer samples with different lattice periods in symmetric and asymmetric RI environments. For the symmetric case, $\lambda_{\max}$ increases continuously with increasing period and asymptotically approaches the $(0, \pm 1)$ RA (red solid line) for the higher periods. These resonances correspond to SLRs for periods of 388 nm and larger (full circles). For smaller periods, the values of $\lambda_{\max}$ are close to the uncoupled LSPR (red horizontal bar) while showing a weak blueshift for decreasing periods. This

blueshift is also observed in the asymmetric RI samples for periods in a very similar range (below approximately 400 nm).

For larger periods, $\lambda_{\max}$ approaches the uncoupled LSPR (gray horizontal bar), indicating that diffractive coupling is not supported due to the strong RI mismatch between the substrate and superstrate in the asymmetric samples. The blueshift observed for decreasing periods is attributed to long-range dipolar interactions. We now want to focus on the SLRs observed in the symmetric RI samples and use the Q -factor to discuss the coupling strength in our self-assembled monolayers. Figure 6d compares the resonance position and Q -factor for the symmetric RI samples as a function of the spectral difference between the diffractive mode and the single particle LSPR ($\Delta\lambda = \lambda_{\text{RA}} - \lambda_{\text{LSPR}}$). Both $\lambda_{\max}$ and the Q -factor decrease in a nearly linear manner for decreasing $\Delta\lambda$ below -50 nm. In this region, diffractive–plasmonic coupling is inhibited and only long-range dipolar interactions become relevant. For positive values of $\Delta\lambda$ or negative values approaching 0 where SLRs appear, the resonance shift becomes steeper and the Q -factor increases significantly, reaching a maximum for $\Delta\lambda \approx 20$ nm corresponding to a period of 418 nm. Here, the Q -factor reaches a value of approximately 20.

For further increases in $\Delta\lambda$, the Q -factor decreases significantly, indicating increasing losses. This decrease is contrary to the observation from simulation in Figure 6a and studies reported in literature where an increase in Q -factor is observed due to the increasing photonic character and reducing plasmonic losses for larger periods. We attribute our findings to the finite domain size in our lattices and scattering losses from domain boundaries and lattice defects. To support this, we compare the resonance FWHM for various domain sizes as a function of lattice period from FDTD simulation in Figure 6e. For domain sizes of two to five, we observe an increase in FWHM with increasing period, while for each period, the larger the FWHM, the smaller the period. In other words, the resonance linewidth increases and thus the Q -factor decreases with decreasing domain size and increasing period. The respective representation of the Q -factor can be found in the Supporting Information (Figure S9). In contrast, the simulation using periodic boundary conditions (and thus resembling an infinite lattice) reveals a decrease in FWHM with increasing period and an increase in Q -factor. Similar effects of the array size on the Q -factor of the SLR were found by Rodriguez *et al.*²⁷

To summarize, there are pronounced differences in the behavior of SLRs in asymmetric and symmetric RI samples. We find the following: (1) A close match between the RI of the substrate and superstrate is required to enable plasmonic coupling to in-plane modes of diffraction. (2) SLRs are only supported for diffraction wavelengths close to, or larger than, the single particle LSPR wavelength. (3) For diffraction wavelengths significantly smaller than the LSPR wavelength, long-range dipolar interactions are observed, leading to a blueshift with decreasing lattice period. (4) High Q -factor SLRs are observed for slightly longer wavelengths of diffraction compared to the LSPR, whereas a reduction in Q -factor is observed for larger offsets of diffraction and LSPR wavelengths. The latter reduction in Q -factor is caused by finite domain size effects and scattering losses.

Despite the obvious influence of lattice imperfections in our self-assembled lattices, the maximum Q -factor of $Q \approx 20$ for a period of 418 nm is more than half of the value obtained from FDTD simulation for an infinite lattice using periodic

boundary conditions ($Q \approx 37$, see Figure S10 in the Supporting Information). Similar calculated results were obtained by Rodríguez *et al.*²⁷ for square arrays of gold disks with 120 nm diameter and 50 nm height. For an infinite array, they calculated $Q \approx 30$ and for finite arrays with a few hundred particles Q already closely approached this limit.

CONCLUSIONS

We have experimentally and theoretically studied the optical properties of periodic gold nanoparticle arrays prepared via interface-mediated self-assembly. While weak electromagnetic coupling related to long-range radiative coupling of the dipoles was observed in strongly asymmetric refractive index environments, homogenization of the refractive index resulted in pronounced surface lattice resonances. In both cases, the coupling strength strongly depends on the interparticle distance. Radiative dipolar interactions increase with decreasing interparticle distance, resulting in a blueshift of the plasmon resonance. In contrast, diffractive–plasmonic coupling becomes pronounced for larger distances where the in-plane modes of diffraction are spectrally close to, or redshifted from, the single particle localized surface plasmon resonance. The spectral overlap between both modes weakens at large enough distances, and then the surface lattice resonances also become weaker in intensity and significantly broaden as a result of the finite size of the crystalline domains in the sample. As a consequence of the decreasing coupling efficiency at increasing lattice constants, the purely plasmonic contribution at lower wavelengths reappears due to energy conservation. Although our experimental samples feature lattice defects and limited single crystalline domain sizes, the deviation of the experimentally observed surface lattice resonance width and intensity from simulations of infinite size lattices is small. This implies that colloidal self-assembly is competitive compared to lithographic approaches commonly applied to the preparation of near-perfect periodic plasmonic lattices. We believe that particle arrays supporting much higher quality factor resonances are readily achievable by self-assembly because the individual plasmonic particle sizes, shapes, and interparticle spacing can all be adjusted.

Our work significantly contributes to a deeper understanding of electromagnetic coupling phenomena in periodic plasmonic arrays. Furthermore, we demonstrate the enormous potential of colloidal self-assembly for the fabrication of cm²-scale samples with tailored plasmonic properties based on relatively simple colloidal building blocks. We believe that our findings can be transferred to other grating-like plasmonic nanostructures such as periodic lines of nanoparticles or ordered supercrystal structures, which might even support collective resonances of higher quality as long as the degree of disorder is low and domain sizes are large. In future work, we will address the possibility for further improvement of the quality factor of surface lattice resonances by optimization of the plasmonic colloidal building blocks and the lattice geometry.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.0c02430>.

Detailed information on colloid synthesis; representative TEM images and UV–Vis spectra from different steps of

synthesis; microstructure analysis of particle monolayers by DFM and AFM; experimental UV–Vis spectra of all particle monolayers and an unordered particle monolayer; local transmission spectra of particle monolayers recorded with an optical light microscope; schematic FDTD simulation layout; calculated Q -factor for various domain sizes; and extinction spectrum from FDTD simulation for the strongest coupling regime (PDF)

Self-assembly process (MP4)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The authors acknowledge financial support by the German Academic Exchange service (DAAD) through its Thematic Network Melbourne-Bayreuth Polymer/Colloid Network sponsored from funds of the Federal Ministry of Education and Research (BMBF). P.M. thanks the ARC for support through Grant CE170100026. M.K. acknowledges the German Research Foundation (DFG) for funding under grant KA3880/6-1. The authors thank Heyou Zhang from the University of Melbourne for his assistance with the DFM measurements and Sophia Köhler from the Heinrich-Heine-University Düsseldorf for her assistance with the TEM measurements.

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